PlugPlan SG - Refined Dialog Map

SC2006 Software Engineering | Team SCE3-04 | Lab 3 | 25 October 2026

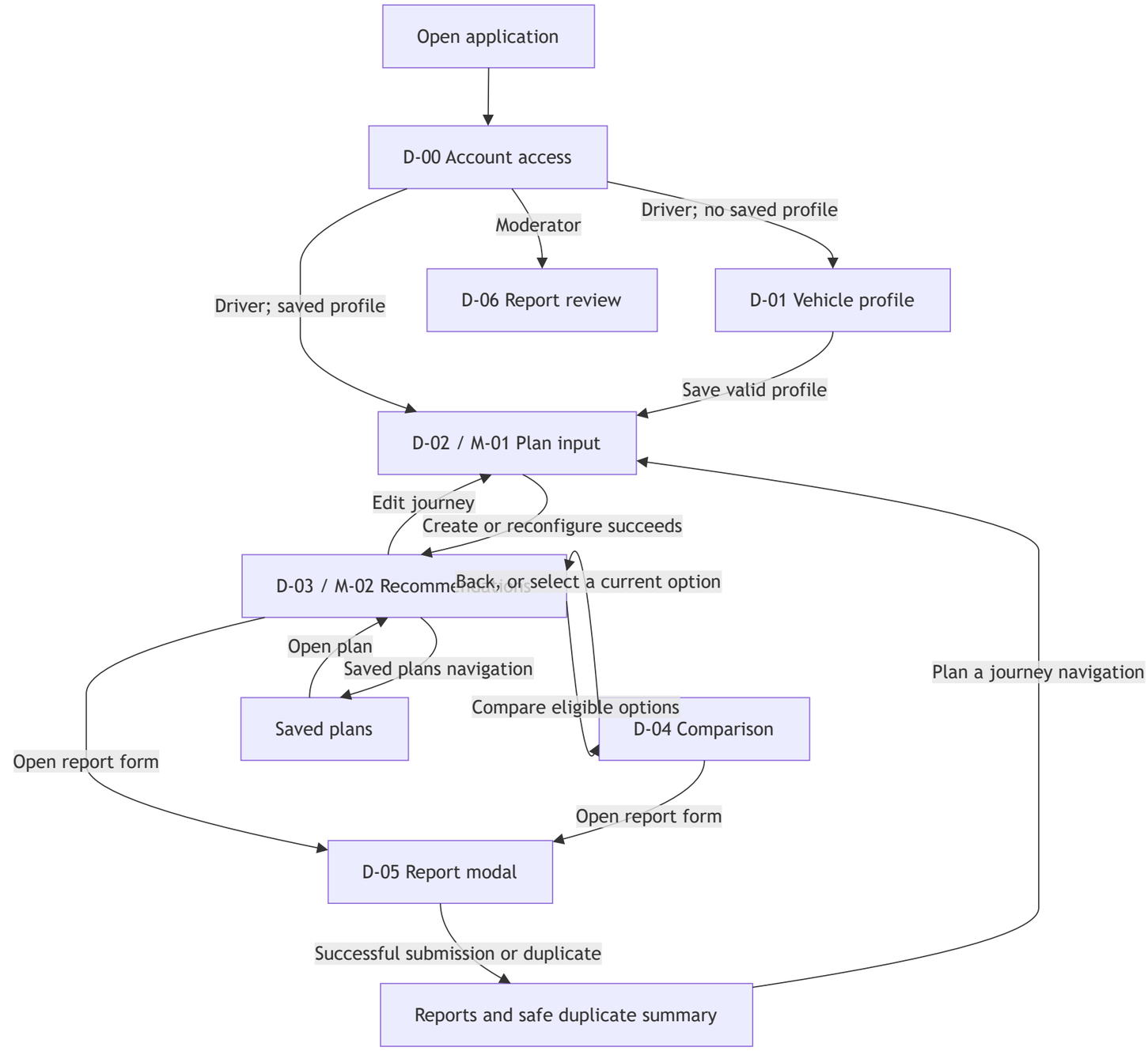
1. Frames and interaction model

The dialog map refines the Lab 2 screen flow into events, guards, actions and recovery states. Screen identifiers remain consistent across the use cases and design model.

Frame	UI state / boundary	Responsibility
D-00	`AuthScreen`	Email/password sign-in and Driver registration
D-01	`view = vehicle`	Create, update or confirm-delete one EV profile
D-02 / M-01	`view = plan`; `PlacePicker`	Journey inputs, explicit Live/Demo mode and reconfiguration
D-03 / M-02	`view = results`; `JourneyMap`	Ranked recommendations, selection, versions and manual refresh
D-04	`view = compare`	Compare two or three options using the same metrics
D-05	`reportContext`; `Modal`	Charger issue form; duplicate summary is a separate overlay
D-06	`view = moderator`; `selectedReport`	Moderator list, evidence, decision form and audit history
Saved plans	`view = saved`	Open a saved plan or confirm-delete it and its versions
Reports	`view = reports`	Driver-owned reports; all reports for a Moderator

M-01 and M-02 are responsive renderings of D-02 and D-03 with the same actions and validation. The mobile workspace uses bottom navigation. Dialogs support keyboard focus and dismissal. Unsaved profile/journey edits trigger a discard confirmation when leaving.

2. Workspace navigation



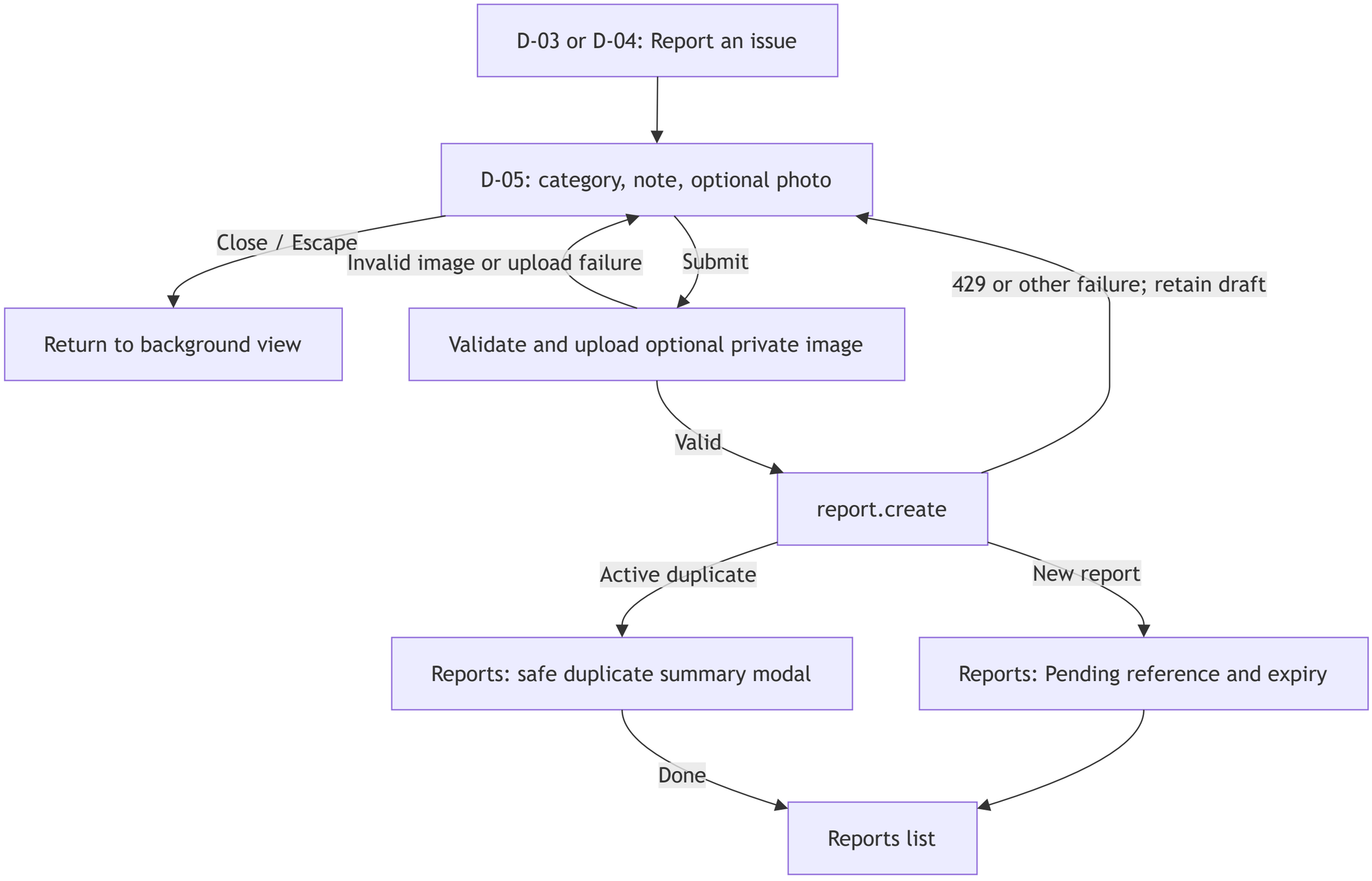
Driver navigation also gives direct access to Vehicle, Plan, Saved plans and Reports. Moderator navigation offers Report review and All reports. A retained session is restored at startup; the persisted `app_users.role` selects the workspace. Sign-out revokes the current application session, signs out through Supabase and returns to D-00.

3. Planning, comparison and refresh transitions

ID	Source and event	Guard / action	Destination
T-01	D-00: sign in / register	Supabase Auth validates credentials; `bootstrap` loads role, profile, plans and reports. Public registration creates a Driver.	D-01, D-02 or D-06 according to role/profile
T-02	Any workspace: sign out	Confirm if edits are unsaved; call `session.logout`, then local SDK sign-out.	D-00
T-03	D-01: save vehicle	Validate field ranges server-side; call `profile.save`; update the saved profile.	D-02
T-04	D-01: delete vehicle	Confirm; call `profile.delete`; reset form defaults. Historical versions retain copied assumptions.	D-01, no active profile
T-05	D-02: search/select place	`places.search` returns candidates in the chosen data mode; selection stores label and coordinates in the draft.	D-02, selected place
T-06	D-02: calculate	Both places selected and a profile saved; validate inputs; call `plan.create`, or `plan.reconfigure` when editing a saved plan.	D-03, newly saved version
T-07	D-02: invalid input / provider failure	Show field or provider error and retain draft; no partial version is stored.	D-02, correction/retry
T-08	D-03: compare	At least two recommendations. With no checkboxes selected, compare all available options; otherwise select two or three.	D-04
T-09	D-03/D-04: choose stop	Viewing current version; call `plan.select`. Update the plan's selection pointer; do not create a version.	D-03, selected option
T-10	D-03: edit / change strategy	Edit copies inputs into D-02; a strategy tab submits `plan.reconfigure` directly.	D-02, or D-03 with new version
T-11	D-03: change version	Load the chosen immutable result; historical views disable selecting a stop.	D-03/D-04, historical evidence
T-12	D-03: refresh	Call `plan.refresh`; recalculate current inputs with latest usable evidence and active verified faults.	D-03, new version
T-13	Refresh: selected option remains eligible	Preserve the selection.	D-03, same choice in new result
T-14	Refresh: selected option disappears	Clear selection, explain the reason and identify the next ranked option. Driver selects explicitly using T-09.	D-03, fallback offered
T-15	Refresh: no fallback / provider failure	No fallback: save an explicit empty result. Failed refresh: preserve the previous version and display the error.	D-03, explanatory state
T-16	Saved plans: open / delete	Open selects the current version; delete requires confirmation and removes the plan and its versions.	D-03 / Saved plans
T-17	Leave a dirty profile/journey	Confirm discarding edits; cancellation keeps the current draft.	Requested view / original view

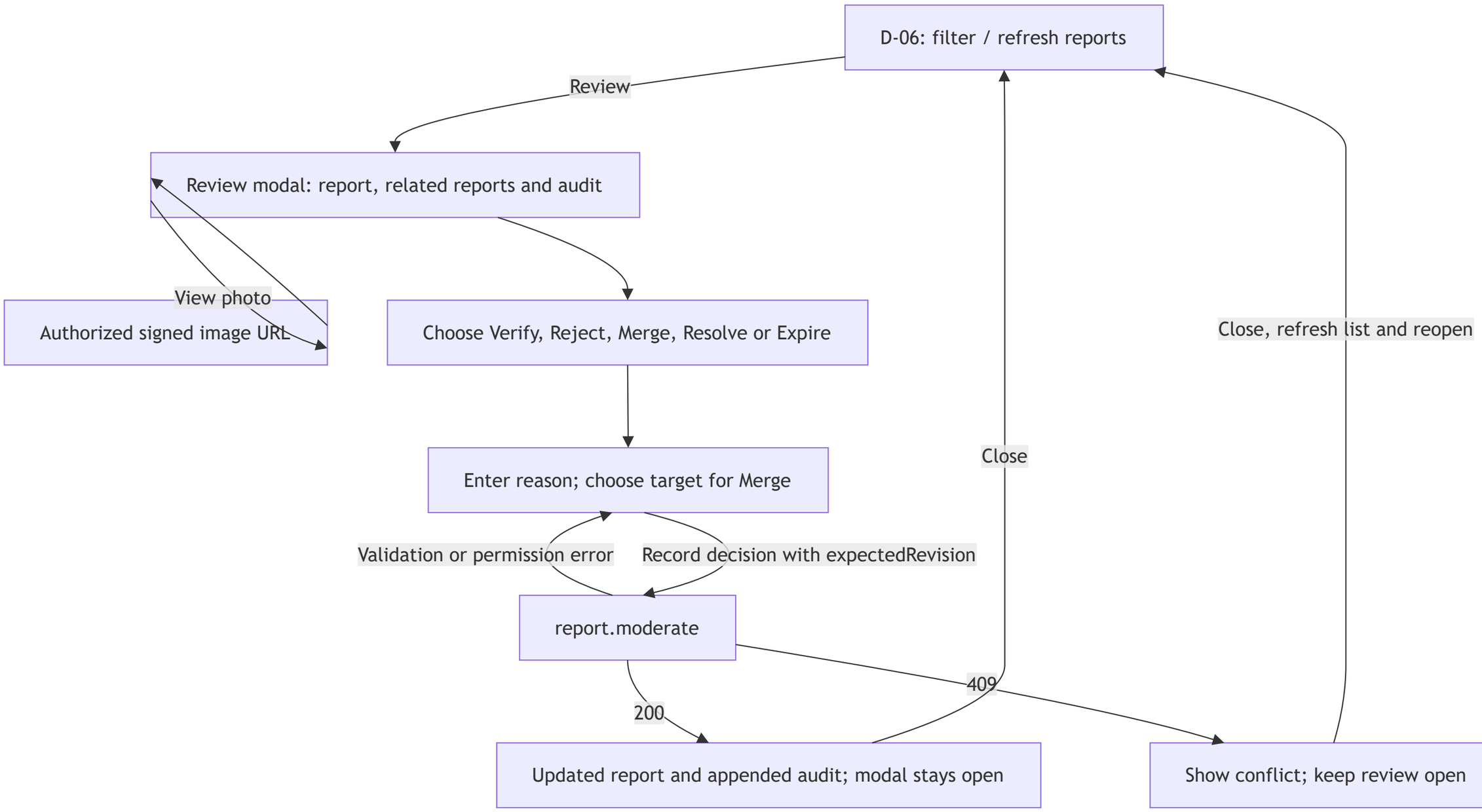
Only creation, reconfiguration and successful refresh append Plan Versions. Selecting a recommendation changes mutable plan state. A successful recalculation ranks feasible candidates, removes stations supported by active verified suspected-fault reports, then forms a shortlist of up to three. Pending reports do not exclude stations.

4. Issue-report dialog



ID	Event / state	Behavior
T-18	Open D-05	Prefill station/connector context; reset note/photo; show privacy guidance and operator support link when known.
T-19	Choose photo	Accept JPEG, PNG or WebP, at most 5 MiB; retain the form when validation/upload fails.
T-20	Submit	UI requires a 5-2,000 character note. Upload any image to the owner folder in private `issue-images`, then call `report.create`.
T-21	New report	Close form, open Reports, acknowledge `PENDING` and default 24-hour expiry.
T-22	Duplicate	Show reference, target, category, status and expiry; another reporter's identity, notes and images remain private.
T-23	Rate limited	Keep draft; display the allowed-again time in SGT from API `retryAfter`.
T-24	View report evidence	`report.image-url` returns a short-lived URL only to an authorized report reader.

5. Moderator dialog



ID	Event / state	Behavior
T-25	Filter / refresh	Filter by report lifecycle state; refreshing calls `reports.list`.
T-26	Review	Inspect note, optional photo, related reports, cached official LTA status and decision history. Official detail retrieval and batch source times are labelled.
T-27	Record decision	Send `reportId`, `moderationAction`, `reason`, `expectedRevision` and optional `mergeTargetId`. UI reason length is 3-500 characters; the server requires a nonblank trimmed reason, at most 500.
T-28	Destructive decision	Reject, Merge, Resolve and Expire request confirmation; Merge requires a valid canonical target.
T-29	Successful decision	Update the report/revision and audit in the open modal; clear the reason field.
T-30	Revision conflict	HTTP 409 leaves the review open with an error. Close, refresh and reopen to review current data before resubmitting.

Only verified, unexpired `SUSPECTED_FAULTY_CONNECTOR` reports support station exclusion. Resolve/expiry removes that report's support; another active support can keep the station excluded. Merge retains the canonical report. These changes affect subsequent calculations and refreshes; saved versions remain historical evidence.

6. UI callouts and design decisions

Area	Controls and constraints
Account access	Email/password form; registration requires at least 10 characters including a letter and number. Busy controls prevent repeat submission. Errors remain generic.
EV profile	Capacity 10-250 kWh; consumption 5-50 kWh/100 km; AC limit 1-50 kW; DC limit 1-400 kW; reserve 5-40% (default 20%); detour 0-120 min; one or more compatible plugs.
Plan input	SOC 0-100%; charging window 5-720 min; detour 0-120 min; optional manual target 0-100%. OneMap suggestions support keyboard selection. Times use SGT unless an explicit offset is supplied.
Strategies	Balanced, Fastest journey, Cheapest charging, Availability first and Minimum detour. Cheapest uses supported total charging cost; Fastest uses total journey time.
Results / comparison	Shared units and assumptions, source and observed/retrieved times, explicit zero/one-result states, unknown price labelled `Price unavailable`, up to three compared options.
Data states	LIVE is fresh fetched evidence; CACHED is usable cached evidence; at 10 minutes or older, relevant source/detail evidence is STALE. DEMO_FIXTURE is explicitly selected synthetic data.
Map	Leaflet uses OneMap tiles and decoded route geometry when supplied; markers represent origin, destination and charging recommendations.
Access control	Browser session uses the Supabase JS client. API requests carry a bearer JWT; `requireUser` checks identity, active session and persisted application role. Driver/Moderator gates and database RLS protect operations.
Persistence	`vehicles`, `plans`, `plan_versions`, `issue_reports` and `moderation_decisions` store owned data; version creation and moderation use transactional RPCs.
Transient data	Unsaved location/profile drafts and modal fields stay in component state. Reload resets unsaved drafts; version inputs are retained only after saving a plan.
Provider failure	Server provider calls share a roughly 3.9-second deadline. Usable station and compatible route caches may support a labelled degraded result; otherwise show an explicit error and retain the draft/previous version.

7. Traceability

Use case	Frames	Main design collaborators
UC-01 Account access	D-00; role-specific landing	AuthScreen, ApiClient, AuthGuard, BootstrapRepository
UC-02 EV profile	D-01	PlugPlanApp, ValidationSchemas, ProfileRepository
UC-03 Create plan	D-02/D-03; M-01/M-02	PlacePicker, ServiceHandler, ProviderGateway, PlanEngine, PlanRepository
UC-04 Review/compare	D-03/D-04; Saved plans	PlugPlanApp, JourneyMap, PlanRepository
UC-05 Reconfigure	D-02/D-03	ServiceHandler, PlanEngine, PlanRepository
UC-06 Refresh/fallback	D-03; M-02	ProviderGateway, PlanEngine, ReportRepository, PlanRepository
UC-07 Issue report	D-05; Reports	PlugPlanApp, Modal, ReportRepository, Supabase Storage
UC-08 Moderation	D-06	PlugPlanApp, AuthGuard, ReportRepository, ModerationDecision